

HealthConnect Clinic — Week 7 HC-POD

Cross-Track Testing & Validation Evidence

Required Documentation

1. Track collaborated with: Data Analytics

2. Project dependency: Per the Week 7 HealthConnect Testing & Validation Matrix: "Data Analytics → Data Science: Validate analytical findings that influence model features or modelling decisions." The Data Analytics track completed its own Week 7 individual testing and, based on those results, directly requested that Data Science validate two specific features using the working model rather than standalone correlation.

3. Component/output being tested: Two Week 6 model features: `long_lead_followup` and `distance_to_clinic_km` (an original dataset field with weak standalone correlation to the outcome).

4. Information/output received (from Data Analytics): Their validated Week 7 findings:

- Booking lead time: 39.13% no-show rate (0–30 days) vs. 63.95% (31–60 days), a +24.82pp gap.
- Previous no-shows: 46.30% (0 prior) vs. 57.83% (≥ 1 prior).
- Distance to clinic: very weak standalone association (Spearman $\rho \approx 0.055$) — Analytics explicitly does not support using distance as a standalone prioritisation factor.
- A Lead Time $\times$ Appointment Type interaction test: Follow-up showed the largest descriptive gap (+28.73pp) but the interaction was **not statistically significant** ($p \approx 0.326$).

Along with this, a direct request: test whether `long_lead_followup` provides incremental out-of-sample predictive value beyond the broader lead-time signal, and whether `distance_to_clinic_km` provides meaningful incremental value in the multivariable model despite its weak standalone correlation.

5. Information/output provided (to Data Analytics): The results of both tests including the reply message summarising findings, and the full notebook with reproducible code.

6. Testing activity completed: A **with-vs-without ablation test** for each feature: the same Logistic Regression pipeline (same patient-grouped split, same preprocessing) trained once with

the feature included and once with it excluded, compared on ROC-AUC (single-split and 5-fold grouped cross-validation), a bootstrap 95% confidence interval on the difference, and — for distance — the top-20%-risk business lift metric.

7. Issue or finding identified:

- **long_lead_followup**: no statistically meaningful incremental value. Single-split AUC difference +0.0017 (bootstrap 95% CI [-0.0008, 0.0041], includes zero); 5-fold CV mean difference +0.0009, winning only 4 of 5 folds. This is consistent with Data Analytics' own finding that the underlying interaction was not statistically significant. Week 6's justification for the feature (Random Forest feature importance $\approx$ 5%) measured how often the model *used* the feature, not how much *unique* signal it added — those turned out to be different things.
- **distance_to_clinic_km**: a small but genuinely consistent incremental contribution. The single-split bootstrap CI was inconclusive ([-0.0089, 0.0096]), but 5-fold cross-validation showed "with distance" beating "without" in **5 out of 5 folds** (mean difference +0.0043), and the top-20%-highest-risk business lift dropped from 1.44x to 1.40x when the feature was removed.

8. Refinement/action taken:

- Removed **long_lead_followup** from the candidate feature set going into Week 8 / the Machine Learning Engineering hand-off — it added complexity without a defensible accuracy benefit.
- Kept **distance_to_clinic_km** in the model — its small multivariable contribution, while modest, is more consistent than its near-zero standalone correlation would suggest.

9. Retest result: For **long_lead_followup**, the "without" run in the ablation test is the retest: performance without the feature is statistically indistinguishable from performance with it (and marginally higher on accuracy/F1 in the single-split comparison). For **distance_to_clinic_km**, the 5-fold cross-validation served as an independent retest of the inconclusive single-split result, and it strengthened rather than weakened the case for keeping the feature.

10. What changed as a result: The Week 8 candidate feature set is now simpler (**long_lead_followup** removed) without any measurable loss of performance, and **distance_to_clinic_km** is confirmed — not just assumed — to belong in the model. Data Analytics also now has a concrete, tested example to cite when explaining to non-technical stakeholders that a weak *standalone* correlation does not automatically mean a feature has no role in a full predictive model.

11. Evidence:

- [HealthConnect_Week7_CrossTrack_DA_DS_Validation.ipynb](#) — full ablation test code, bootstrap significance tests, 5-fold cross-validation, business-lift comparison, and the per-fold chart for distance.

12. How this activity improved the overall HealthConnect solution: This is the clearest instance yet of the Week 7 requirement working as intended: a real request from another track, tested with a defensible method (ablation + significance testing, not opinion), that changed an actual decision — one feature removed, one feature's inclusion confirmed with stronger evidence than before. It also produced a piece of shared understanding (weak standalone correlation $\neq$ no multivariable value) that is useful to both tracks' future work, not just to Data Science's model.